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Combination Toilet and Urinal

Abstract

The present invention introduces a toilet-urinal combination design, in rim and rimless options, encompassing a bowl structure with a semi-circular cup-shaped extension. A water flush design has been shown to clean the bowl extension for both the rim and rimless options. The invention is suitable for manufacture in stainless steel or ceramic materials such as porcelain and vitreous china. The design allows users who urinate standing up to stand closer to the bowl with legs on either side of the protruding cup-shaped extension, reducing the risk of unintentional spills and providing enhanced user comfort during urination. Importantly, the incorporation of this semi-circular cup-shaped extension ensures that genital contact with the bowl is minimized during seated urination and urinating while defecating, adding an extra user comfort and hygiene. The urinal extension design also allows the user better access to self-care when seated.

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Background/Summary

BACKGROUND ON THE INVENTION

[0001] This toilet-urinal combination effectively mitigates the common problem of urine spillage in diverse settings such as homes, commercial use, public venues, portable loos, outdoor concerts, stadiums and mixed gender public lavatory facilities.

[0002] The augmented toilet design contributes to improved health, hygiene, and significant cost reduction associated with frequent cleaning and maintenance.

[0003] The invention addresses common challenges associated with traditional toilets, particularly the issue of urine spillage and discomfort during use. By introducing an extended bowl design with a semi-circular cup-shaped addition, the fixture becomes a dual-purpose toilet-urinal.

[0004] This innovation not only captures and contains urine drips, preventing floor spillage, but also allows users to stand closer to the bowl, with legs either side of the extension, minimizing unintentional spills.

[0005] The design also reduces the risk of genital contact with the bowl and reduces splashes during seated urination and urination while defecating. The urinal extension design also allows the user better access to self-care hygiene while seated.

[0006] A rim and a rimless bowl design with the urinal extension and cleaning flushes for both are presented.

BRIEF SUMMARY OF THE INVENTION

[0007] The invention addresses common challenges associated with existing toilet designs, particularly the issue of urine spillage and discomfort during use. By introducing an extended bowl design with a semi-circular cup-shaped addition, the fixture becomes a dual-purpose toilet-urinal.

[0008] This invention includes a toilet seat and cover to fit the rim and rimless new bowl shape and a different flush mechanism to clean the extended bowl of the rim and rimless bowls.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1: Isometric view presents the overall toilet-urinal hybrid featuring an extended bowl with a semi-circular cup-shaped extension at the front 1. This image shows the toilet-urinal with a rim with concealed flush holes under the rim as shown in FIG. 2.

[0010] This view illustrates the water flush fed from a gravity cistern above the bowl and dotted line detail shows the waste pipe with u-bend to prevent smell coming back up the toilet from the drains. The accompanying seat and lid are not shown in this view.

[0011] FIG. 2: This plan shows the water channel flush detail 2 with flush holes under the toilet rim. This detail shows how the flush water will flow around the bowl and the extension 3 cleaning the bowl and extension through small flush holes all-round the channel which goes completely round the bowl and extension.

[0012] This plan also shows the position of the section views taken across AA and BB which are shown in FIGS. 3 and 4 respectively.

[0013] FIG. 3: An enlarged sectional detail plan of the bowl through BB of the semi-circular cup-shaped extension 4 highlights its strategic position for capturing and containing urine drips. The flush detail 5 shows how the flush water will be routed around the bowl extension. The extension length 6, is shown as the same thickness as the toilet bowl but this can be slightly longer or slightly shorter based on the shape of the bowl.

[0014] The dimensions can be adjusted depending on the size and shape of the toilet bowl to make it both proportionate, practical and aesthetically pleasing.

[0015] FIG. 4: A sectional view of the bowl through AA reveals the integrated urinal feature within the extended bowl, with the sectional detail of the bowl flush 7 which goes around the bowl extension. The urinal extension is cleaned by the normal toilet flush from the water cistern or other

water supply, as shown by the two alternative option arrows **8**, and the flush water flow round to clean the urinal extension.

[0016] The shaped seat **9** and lid **10** are tailored to fit the new bowl extension design.

[0017] The toilet pedestal base or alternative wall fixings, **11**, should be adjusted to allow for the cantilever impact of the extension and be designed to accommodate the weight of anyone sitting on the edge of the toilet seat.

[0018] FIG. **5**: This plan view showcases an alternative rimless extended bowl **12**. This structure can be manufactured in all common ceramic materials however it is ideal for alternative material construction such as stainless steel.

[0019] The bowl plan shape is similar to FIGS. **1** in that the design allows the optimized user proximity during urination as the shape allows the legs to get closer to the bowl extension to catch urine spills.

[0020] The view also shows the flush pipe detail **13** which carries the water from the cistern or other water source to the flush deflector **14** which directs the flush around the bowl. There is an additional deflector for the bowl extension **15** which deflects the flush water into the bowl extension to clean the bowl.

[0021] The accompanying seat and lid are not shown in this view.

[0022] This plan also shows the position of the section views taken across AA and BB which are shown in FIGS. **6** and **7** respectively.

[0023] FIG. **6**: A sectional view of the rimless bowl through AA reveals the integrated urinal feature within the extended bowl, with an additional detail of the main bowl flush deflector **16** and the extension bowl deflector **17**.

[0024] The urinal extension is cleaned by the normal toilet flush from the water cistern or other water supply, as shown by the two alternative option arrows, and the deflector directs flush water flow to clean the urinal extension.

[0025] The shaped seat **18** and lid **19** are tailored to fit the bowl extension design.

[0026] The toilet pedestal base or alternative wall fixings, **20**, should be adjusted to allow for the cantilever impact of the extension and be designed to accommodate the weight of anyone sitting on the edge of the toilet seat.

[0027] FIG. **7**: An enlarged sectional detail plan view of the rimless bowl through BB of the semi-circular cup-shaped extension **21** highlights its strategic position for capturing and containing urine drips.

[0028] The flush deflector **22** shows how the flush water will be redirected by the deflector around the bowl extension.

[0029] The extension length **23**, is shown as the same thickness as the toilet bowl but this can be can be slightly longer or slightly smaller based on the shape of the bowl. The dimensions can be adjusted depending on the size and shape of the toilet bowl to make it both proportionate, practical and aesthetically pleasing.

[0030] FIG. **8**: The accompanying seat is uniquely shaped to align seamlessly with the shape contours of the new bowl design and fit the rim of the bowl and provide support. The two fixings to hold the seat via the fixing holes in the bowl are shown as **24**.

[0031] FIG. **9**: The lid is shaped to completely cover the bowl when toilet not in use. The same fixings mentioned in FIG. **8** hold the lid and seat to the bowl are shown as **25**.

[0032] SUMMARY: These detailed drawings provide a visual narrative of the combined toilet and urinal invention, showcasing both the rim and rimless options with alternative flush systems for enhanced cleaning of the urinal extension along with the integration of shaped seat and lid elements.

DETAILED DESCRIPTION OF THE INVENTION

[0033] The invention resides in the technical field of sanitary fixtures and plumbing infrastructure, specifically, toilet bowl and urinal design with a semi-circular cup-shaped addition to create a dual-

purpose toilet-urinal.

[0034] The invention introduces an enhancement to traditional toilet fixtures addressing longstanding challenges in hygiene and user comfort. By extending the bowl design with a semi-circular cup-shaped extension, the invention creates a dual-purpose toilet-urinal.

[0035] This innovative approach not only effectively captures and contains urine drips, preventing floor spillage, but also facilitates a more comfortable user experience.

[0036] The invention enables users who stand to urinate, to stand closer to the bowl with legs on either side of the protruding cup-shaped extension, reducing the risk of unintentional spills during urination. Additionally, the design minimizes the chance of genital contact with the bowl and splashes during seated urination and urination while defecating adding an extra layer of user comfort and hygiene.

[0037] The urinal extension design also allows the user better access to self-care hygiene when seated.

[0038] Versatility is a key aspect of the invention, offering the option for manufacturing in stainless steel or traditional ceramic materials to suit diverse preferences. The integrated urinal feature contributes to improved health and hygiene, while the design's compatibility with existing plumbing infrastructure ensures seamless replacement.

[0039] Prior art primarily encompasses traditional toilet designs typically with standard oval bowl configurations. This art includes various attempts to address these issues, such as anti-splash technologies and modifications to the bowl shape. However, these solutions typically focus on mitigating urine splash from the bowl itself rather than preventing the more frequent spillage onto the floor.

[0040] Prior art also includes separate urinals in public facilities, necessitating dedicated spaces and plumbing. These solutions are often impractical in mixed-gender facilities, which are becoming increasingly common, or residential settings, where space constraints and additional plumbing complexities pose extra cost.

Embodiment 1: Integrated Design with Shaped Seat and Lid

[0041] This embodiment features an enhanced toilet-urinal hybrid with an extended bowl and a semi-circular cup-shaped extension at the front. The seat and lid are uniquely shaped to match the contours of the extended bowl, providing a seamless and ergonomic fit.

[0042] This design ensures optimal user comfort and minimises genital contact with the bowl and splashes during both seated urination and urination while defecating. The urinal extension design also allows the user better access for self-care hygiene when seated.

Embodiment 2: Proximity-Optimized Configuration

[0043] In this embodiment, the semi-circular cup-shaped extension is strategically positioned to enable users, who urinate standing, to stand closer to the bowl with legs on either side of the bowl. This embodiment prioritizes user comfort and addresses common spillage challenges associated with conventional toilet designs.

Embodiment 3: Cleaning-Enhanced Flush System

[0044] This embodiment for the rimless toilet urinal combination includes a two-deflector flush system with a second deflector at the start urinal extension to deflect the flush water around the bowl extension. This design ensures thorough cleaning in the urinal extension area, contributing to improved hygiene.

Embodiment 4: Material Adaptability

[0045] A variant embodiment emphasizes the adaptability of the invention with alternative manufacturing materials and a choice of rim or rimless options. The invention can be manufactured in stainless steel or traditional ceramic, providing flexibility to suit diverse settings such as homes, public venues, commercial use, portable loos, outdoor concerts, stadiums and mixed gender public lavatory facilities to effectively mitigate the common problem of urine spillage.

Claims

- 1: COMBINED TOILET AND URINAL** A toilet-urinal fixture comprising a protruding semi-circular cup-shape at the front, transforming the fixture into a versatile unit serving both as a traditional toilet and a urinal; the extended bowl configuration is adept at capturing and containing urine drips, effectively preventing spillage onto the floor; this design ensures optimal user comfort and minimizes genital contact with the bowl and splashes during both seated urination and urination while defecating; the urinal extension design also allows the user better access to self-care hygiene when seated.
- 2: PROXIMITY-OPTIMISED URINAL CONFIGURATION** The toilet-urinal combination fixture of claim 1 comprising the design of the protruding semi-circular cup-shape extension facilitates users to stand in closer proximity to the bowl positioning legs on either side; this optimized configuration not only enhances user comfort but also minimizes the risk of inadvertent spills during urination.
- 3: HYGENIC URINE CONTAINMENT** The toilet-urinal fixture of claim 1, has flush options for rim and rimless construction comprising: for the rim option holes under the rim which encircle the bowl and extension ensures the flush water flushes both the bowl and extension; for the rimless option a special deflector near the bowl extension ensures the flush water is deflected which ensures the water flush encircles the urinal extension of the bowl; and both options ensure efficient removal of urine, contributing to elevated hygiene and cleanliness standards.
- 4: ADAPTABILITY TO EXISTING PLUMBING INFRASTRUCTURE** The toilet-urinal fixture of claim 1, comprising a design for seamless integration into established plumbing infrastructure; the design facilitates retrofitting into conventional lavatory setups without necessitating extensive modifications.
- 5: COST-EFFECTIVE CLEANING SOLUTION** A method for reducing cleaning and maintenance costs in toilet facilities, leveraging the features of the toilet-urinal fixture as set out in claim 1; comprising the innovative design minimizes urine spillage on the floor, subsequently decreasing the frequency and intensity of cleaning operations, leading to a cost-effective and efficient solution.
- 6: VERSATILE APPLICATION** The toilet-urinal fixture of claim 1, in rim and rimless options, is adaptable for deployment in diverse settings comprising: homes, commercial use, public venues, portable loos, outdoor concerts, stadiums and mixed gender public lavatory facilities and the invention can be manufactured using stainless steel, porcelain or ceramic where this adaptability provides a comprehensive and versatile solution to enhance overall sanitation.
- 7: NON-OBVIOUS DESIGN** The toilet-urinal fixture of claim 1 wherein the incorporation of the semi-circular cup-shaped extension represents a non-obvious change to conventional toilet designs; this unique and innovative design provides an ideal solution to the perennial issue of urine spillage, concurrently optimizing the user experience and advancing the state of the art in toilet technology.
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